

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

May - June 2023

20BTC103 – BIOLOGY OF STEM CELLS

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – III each** and **One full** question from **Unit – II**.

2) Draw diagrams when necessary.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Give an account on the culturing and markers of embryonic stem cells.	10	L*1	1	1.2.1
	b) Write short notes on: i) Cardiac stem cells, ii) Adult stem cells.	5+5	L1	1	1.2.1
2.	a) List the uniqueness and usefulness of different types of stem Cells.	10	L1	1	1.2.1
	b) Give a comparative account of Embryonic stem cells Vs Mesenchymal stem cells.	10	L1	2	1.2.1
3.	a) Enlist the characteristics of Stem cells in the adult kidney. Highlight their usefulness in therapy for renal failure.	10	L1	2	1.2.1
	b) Write short notes on: i) Liver stem cells, ii) Amniotic fluid-derived pluripotent cells.	5+5	L1	2	1.2.1
Unit – II					
4.	a) Outline the usefulness of hematopoietic stem cells in cancer and immunotherapy.	10	L1	3	1.2.1
	b) Summarize the isolation of hematopoietic stem cells based on function and biological response.	10	L1	3	1.2.1
5.	a) Elaborate on the potency of Hematopoietic stem cell transplantation for treating autoimmune diseases.	10	L1	4	1.2.1
	b) Discuss in detail the role of hematopoietic stem cells in the treatment of solid tumors.	10	L1	4	1.2.1
Unit – III					
6.	a) Discuss in detail the repair of myocardial damage by stem cells.	10	L1	5	1.2.1
	b) Outline the manipulation of stem cells for cell replacement in the Central Nervous System.	10	L1	5	1.2.1
7.	a) Elaborate on the characteristics, cultivation, and transplantation of keratinocyte stem cells.	10	L1	5	2.1.2
	b) Write short note on: i) Ex-vivo reconstruction of cells and scaffolds; ii) Stem cells for ligaments and tendon repair.	5+5	L1	5	1.2.1
8.	a) Discuss the role of stem cells in tissue engineering.	10	L1	5	1
	b) Elaborate on the orthopedic application of stem cells.	10	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
